Task 1 | Technical use guide

How to run and inspect the prototype | 17 October 2026

Start

Install Docker Desktop or Docker Engine with Docker Compose. Clone the repository after it is created, open a terminal in the project folder, then run:

```
docker compose up --build
```

The first run downloads the UCI dataset and can take several minutes. The loader exits after ingestion; MongoDB remains running and keeps records in the `mongo_data` volume. Run `docker compose down` to stop while preserving data, or `docker compose down -v` to remove the volume and reset the database.

Inspect the result

Open a second terminal and connect with the educational credentials shown in Compose:

```
docker compose exec mongo mongosh --username root --password local-only-change-me
--authenticationDatabase admin city_air
```

```
db.measurements.countDocuments({})
db.measurements.findOne({station: "Aotizhongxin"})
```

One document represents one station-hour. It contains `_id`, station, BSON timestamp, source timezone, source, and measurements. Missing CSV values are omitted. The compound index supports station and time filters. Configuration values for batch size and retry limits are in `docker-compose.yml`.

Limitations before submission

The source must be reachable when ingestion starts. The local credentials are for learning only. Docker was unavailable in the authoring environment, so the end-to-end database run has not yet been verified; run it and record the actual document count. No GitHub repository has been published from this workspace. Replace the placeholder URL in the README and phase documents after pushing the code.

Repository URL to add: https://github.com//municipal-air-quality-batch-ingestion

Source: UCI Machine Learning Repository, Beijing Multi-Site Air Quality, https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data